

Cohomology and flabby sheaves

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1 A category introduction

1.1 Abelian Category

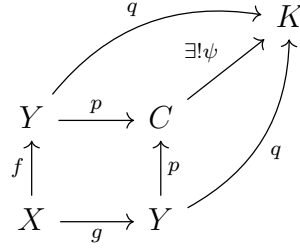
Definition 1.1. Let $\mathcal{C}$ be a category and let $X, Y \in \text{Ob}(\mathcal{C})$ be two objects and let $f, g \in \text{Hom}_{\mathcal{C}}(X, Y)$ be two morphisms. An *equalizer* of the pair (f, g) is a pair $\text{Eq}(f, g) := (C, p)$ with $C \in \text{Ob}(\mathcal{C})$ and $p \in \text{Hom}_{\mathcal{C}}(C, X)$ such that $f \circ p = g \circ p$

$$\begin{array}{ccc} X & \xrightleftharpoons[f]{g} & Y \\ & \nwarrow p & \\ & & C \end{array}$$

and such that for any couple (K, q) with $K \in \text{Ob}(\mathcal{C})$ and $q \in \text{Hom}_{\mathcal{C}}(K, X)$ there exists a unique morphism $\psi \in \text{Hom}_{\mathcal{C}}(K, C)$ such that $p \circ \psi = q$; i.e. we require that the following diagram toggles

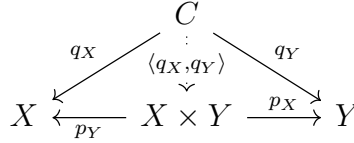
$$\begin{array}{ccccc} & & X & \xrightarrow{f} & Y \\ & \nearrow q & \uparrow p & & \uparrow g \\ & & C & \xrightarrow{p} & X \\ & \nwarrow \exists! \psi & & \nearrow q & \\ K & & & & \end{array}$$

One can also define the *coequalizer* reversing the arrows, i.e



and we will say that the $\text{Coeq}(f, g) = (C, p)$.

Definition 1.2. The *(binary) product* of two objects X, Y in a category $\mathcal{C}$ is a tern $(X \times Y, p_X, p_Y)$ of an object $X \times Y \in \mathcal{C}$ and morphisms $p_X : X \times Y \rightarrow X$ and $p_Y : X \times Y \rightarrow Y$ such that for every similar term $(C, q_X : C \rightarrow X, q_Y : C \rightarrow Y)$ there is a unique morphism $\langle q_X, q_Y \rangle := \Delta(q_X, q_Y)$ (the diagonal morphism) that makes commutative the following diagram:



We say that a category *has binary products* if any pair of objects in the category has binary product.

Reversing the arrows in the previous definition we get the definition for the *coproduct* of two objects.

Definition 1.3. Let $\mathcal{C}$ a pointed category and let $X, Y \in \text{Ob}(\mathcal{C})$. Let's consider $\text{Hom}_{\mathcal{C}}(X, 0) = \{!^X\}$ and $\text{Hom}_{\mathcal{C}}(0, Y) = \{!_Y\}$; then the *zero morphism* $0_{X,Y} \in \text{Hom}_{\mathcal{C}}(X, Y)$ is the morphism

$$X \xrightarrow{\{!^X\}} 0 \xrightarrow{\{!_Y\}} Y$$

The above morphism will be simply called $0 : X \rightarrow Y$

Definition 1.4. Let $f : X \rightarrow Y$ be a morphism in a pointed category; a *kernel* of f is an equalizer of the pair $(f, 0)$, i.e. $\text{Ker}(f) := \text{Eq}(f, 0)$

Definition 1.5. Let $f : X \rightarrow Y$ be a morphism in a pointed category; a *cokernel* of f is a coequalizer of the pair $(f, 0)$, i.e. $\text{Coker}(f) := \text{Coeq}(f, 0)$.

Definition 1.6. A monomorphism in a category $\mathcal{C}$ is a morphism $m : B \rightarrow C$ such that if $m \circ f = m \circ g$, then $f = g$ for any parallel pair of morphisms $f, g \in \text{Hom}_{\mathcal{C}}(A, B)$

reversing the arrows we obtain the definition of *epimorphism*.

Definition 1.7. An **abelian category** is a category $\mathcal{A}$ such that:

- $\mathcal{A}$ has a null object, i.e. an object such that is an initial object and a final object;

- for any $A, B \in \text{Ob}(\mathcal{A})$, $\text{Hom}(A, B)$ is an abelian group;
- It has binary products and binary coproducts;
- every morphism has a kernel and a co-kernel;
- Every mono in $\mathcal{C}$ is a kernel and every epi in $\mathcal{C}$ is a cokernel.

1.2 Complexes

Definition 1.8. definizione di successione per oggetti di categorie. (magfari da mettere più in basso) Let $\mathcal{C}$ be an abelian category; a *complex of object in $\mathcal{C}$* is a sequence of objects $(X_i)_{i \in \mathbb{N}}$ and morphisms $(f^i : X^i \rightarrow X^{i+1})_{i \in \mathbb{N}}$ such that $f^{i+1} \circ f^i = 0_{X^i, X^{i+2}}$. Given a complex $(X^i, f^i)_{i \in \mathbb{N}}$ of object in $\mathcal{C}$, we will say that is an *exact sequence* if it holds that

Definition 1.9. Let $\mathcal{C}$ be an abelian category, $A, B, C \in \text{Ob}(\mathcal{C})$ objects and $f : A \rightarrow B$, $g : B \rightarrow C$ morphisms with $g \circ f = 0_{A, C}$, so we have the following diagram:

$$A \xrightarrow{f} B \xrightarrow{g} C$$

1. The above sequence is called *kernel-exact* in B if for every kernel $(K, j : K \rightarrow B)$ of g , the unique morphism $\bar{f} : A \rightarrow K$ such that $j \circ \bar{f} = f$ is an epimorphism.

$$\begin{array}{ccc} A & \xrightarrow{f} & B \xrightarrow{g} C \\ \exists! \bar{f} \downarrow & \nearrow j & \\ K & & \end{array}$$

2. The above sequence is called *cokernel-exact* in B if for every kernel $(D, p : B \rightarrow D)$ of g , the unique morphism $\bar{g} : A \rightarrow D$ such that $\bar{g} \circ p = f$ is a monomorphism.

$$\begin{array}{ccc} A & \xrightarrow{f} & B \xrightarrow{g} C \\ & \searrow p & \uparrow \exists! \bar{g} \\ & & D \end{array}$$

Theorem 1.10. The above definitions are equivalent.

Definition 1.11. A complex $(X^i, f^i)_{i \in \mathbb{N}}$ of object in $\mathcal{C}$ is called *exact in $i + 1$* if the sequence

$$X^i \rightarrow X^{i+1} \rightarrow X^{i+2}$$

is kernel-exact. A *resolution* for an object $X \in \text{Ob}(\mathcal{C})$, if it exists, is a complex $(X^i, f^i)_{i \in \mathbb{N}}$ with a morphism $\epsilon : X \rightarrow X^0$ such that the sequence

$$0 \rightarrow X \rightarrow X^0 \rightarrow X^1 \rightarrow \dots$$

is exact in each i .

1.3 Injectives and derived functors

Definition 1.12. An object $Q \in \text{Ob}(\mathcal{C})$ for a category $\mathcal{C}$ is said to be *injective* if for any monomorphism $f : X \rightarrow Y$ and every morphism $g : X \rightarrow Q$ there exists a morphism $h : Y \rightarrow Q$ such that $h \circ f = g$, i.e. such that the following diagram toggles

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \downarrow g & \nearrow h & \\ Q & & \end{array}$$

Definition 1.13. Let $\mathcal{C}$ be a category; we say that an object $A \in \text{Ob}(\mathcal{C})$ can be embedded in an object $B \in \text{Ob}(\mathcal{C})$ if there exists a monomorphism $f : A \rightarrow B$.

Definition 1.14. Let $\mathcal{C}$ be a category. We say that $\mathcal{C}$ has enough *injectives* if every object can be embedded in an injective object.

Definition 1.15. Let $\mathcal{C}$ be a category. An *injective resolution* for an object $A \in \text{Ob}(\mathcal{C})$ is a resolution

$$I^\bullet : 0 \rightarrow I^0 \rightarrow I^1 \rightarrow \dots$$

for A where for any $i \in \mathbb{Z}_{\geq 0}$, I^i is an injective object.

Remark 1.16. Let $\mathcal{C}$ be a category such that it has enough injectives, then it is clear that for any object A there exists an injective resolution. Indeed there exists an injective object I and a monomorphism $f \in \text{Hom}_{\mathcal{C}}(A, I)$, then we can build $K = \text{Coker}(f)$ and this is again an object in $\mathcal{C}$, hence there exists I^1 an injective object and a monomorphism $f^1 \in \text{Hom}_{\mathcal{C}}(K, I^1)$; then we have

$$0 \rightarrow A \rightarrow I \rightarrow I^1 \rightarrow \dots$$

where the map $I \rightarrow I^1$ is the composition $f^1 \circ \text{Coker}(f)$.

Now let us consider an abelian category $\mathcal{C}$ with enough injectives and a left exact functor $^1 F : \mathcal{C} \rightarrow \mathcal{D}$ to another abelian category $\mathcal{D}$. We can define the i -th right derived functor $R^i F$ of F by choosing an injective resolution $I^\bullet$ for each object A and setting

$$R^i F(A) := H^i(F(I^\bullet))^2$$

One can prove also that the definition does not depend on $I^\bullet$. Moreover, given a short exact sequence

$$0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$$

of object in $\mathcal{C}$, the functor F induces a long exact sequence

$$0 \rightarrow R^0 F(A) \rightarrow R^0 F(B) \rightarrow R^0 F(C) \rightarrow R^1 F(A) \rightarrow R^1 F(B) \rightarrow R^1 F(C) \rightarrow \dots$$

One can also prove that $R^0 F \cong F$ and that $R^i F(I) = 0$ for $i > 0$ and I injective ³.

¹A *left exact functor* is a functor such that if $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ is kernel-exact sequence then $0 \rightarrow F(A) \rightarrow F(B) \rightarrow F(C)$ is a kernel exact sequence.

²Given a complex $A \xrightarrow{f} B \xrightarrow{g} C$, we can define first of all $\text{Im} f := \text{Ker}(\text{Coker} f)$. Then one can prove also that there exists a unique monomorphism $h : \text{Im} f \rightarrow \text{Ker} g$. In the end we can define $H(B) := \text{Coker} h$

³Since we have that $0 \rightarrow I \rightarrow I \rightarrow 0 \rightarrow 0 \rightarrow \dots$ is an injective resolution for I .

Remark 1.17. One can also prove that any abelian category admits an equivalence of category with a subcategory of the A – modules⁴. In this way, one could skip the above discussion and define all the previous object through the functor.

⁴There exists a faithful ("injective"), full ("surjective") and exact functor between the category.

2 Flabby sheaves and cohomology

2.1 Flabby sheaves

Lemma 2.1. The category of sheaves of abelian groups on a topological space X has enough injectives.

By the lemma we can define

$$H^i(X, \mathcal{F}) := R^i\Gamma(X, \mathcal{F})$$

where $\Gamma(X, -)$ is the left exact functor

$$\begin{array}{ccc} \Gamma(X, -) : \text{Sh}_{\text{AbGrp}, X} & \rightarrow & \text{AbGrp} \\ \mathcal{F} & \mapsto & \mathcal{F}(X) \end{array}$$

Definition 2.2. Let X be a topological space and $\mathcal{F}$ a sheaf on X . We will say that $\mathcal{F}$ is a **flabby** sheaf if for any $U \subset X$ open set the restriction:

$$\rho_U^X : \mathcal{F}(X) \rightarrow \mathcal{F}(U)$$

is surjective.

Definition 2.3. Let X a topological space; for any $U \subset X$ open subset we have a sheaf $\mathbb{Z}_U$ defined in the following way: let $V \subset X$ an open set, then if $V \subset U$ we have $\mathbb{Z}_U(V) = \mathbb{Z}$ otherwise $\mathbb{Z}_U(V) = \{0\}$. If $A \subset B$ The restriction's maps are:

- if $B \subset U$ is the identity;
- otherwise is the zero map.

Lemma 2.4. For a sheaf $\mathcal{F}$ on X there are isomorphisms of abelian groups

$$\text{Hom}(\mathbb{Z}_U, \mathcal{F}) \cong \mathcal{F}(U)$$

for any $U \subset X$ open subset.

Proof. Let's define the following function:

$$\begin{array}{ccc} a : \text{Hom}(\mathbb{Z}_U, \mathcal{F}) & \rightarrow & \mathcal{F}(U) \\ f & \mapsto & f(U)(1) \end{array}$$

it's easy to check that the a is a linear map. Moreover we have the function

$$\begin{array}{ccc} a^{-1} : \mathcal{F}(U) & \rightarrow & \text{Hom}(\mathbb{Z}_U, \mathcal{F}) \\ x & \mapsto & g_x : \mathbb{Z}_U \rightarrow \mathcal{F} \end{array}$$

where for any $V \subset X$ open set we define:

- if $V \subset U$:

$$\begin{array}{ccc} g_x(V) : \mathbb{Z}_U(V) & \rightarrow & \mathcal{F}(V) \\ 1 & \mapsto & \rho_V^U(x) \end{array}$$

- if $V \not\subset U$:

$$\begin{array}{ccc} g_x(V) : \mathbb{Z}_U(V) & \rightarrow & \mathcal{F}(V) \\ 1 & \mapsto & 0 \end{array}$$

And it is easy to check that $a \circ a^{-1} = \text{id}_{\mathcal{F}(U)}$ and $a^{-1} \circ a = \text{id}_{\text{Hom}(\mathbb{Z}_U, \mathcal{F})}$ □

Proposition 2.5. Every injective sheaf is flabby.

Proof. Let $\mathcal{I}$ be an injective sheaf, $U \subset X$ be an open set and be $x \in \mathcal{I}(U)$. From the previous lemma we can consider $a^{-1}(x) : \mathbb{Z}_U \rightarrow \mathcal{I}$ as a morphism of sheaves. Now we can consider $\mathbb{Z}_X$ and let's define the morphism $\psi : \mathbb{Z}_U \rightarrow \mathbb{Z}_X$ of sheaves defined as follows: given an open set $V \subset X$:

- if $V \subset U$: $\psi(V) := \text{id}_{\mathbb{Z}}$
- if $V \not\subset U$: $\psi(V) := 0$

Since $\mathcal{I}$ is an injective object and since ψ is a monomorphism then there exists another morphism $h : \mathbb{Z}_X \rightarrow \mathcal{I}$ such that the following diagram commutes

$$\begin{array}{ccc} \mathbb{Z}_U & \xrightarrow{\psi} & \mathbb{Z}_X \\ a^{-1}(x) \downarrow & \swarrow h & \\ \mathcal{I} & & \end{array}$$

then we have that $h \in \text{Hom}(\mathbb{Z}_X, \mathcal{I})$. Since h is a morphism of sheaves, we have that the following diagram

$$\begin{array}{ccc} \mathbb{Z}_X(X) & \xrightarrow{\text{id}_{\mathbb{Z}}} & \mathbb{Z}_X(U) \\ h(X) \downarrow & & \downarrow h(U) \\ \mathcal{I}(X) & \xrightarrow{\rho_U^X} & \mathcal{I}(U) \end{array}$$

commutes, then we have

$$\rho_U^X(h(X)(1)) = h(U)(\text{id}_{\mathbb{Z}}(1)) = h(U)(\psi(U)(1)) = a^{-1}(x)(U)(1) = g_x(U)(1) = \rho_U^U(x) = x$$

Thus $\mathcal{I}$ is flabby. □

2.2 Cohomology of sheaves

Lemma 2.6. Let

$$0 \longrightarrow \mathcal{F} \xrightarrow{\varphi} \mathcal{G} \xrightarrow{\psi} \mathcal{H} \longrightarrow 0$$

be a short exact sequence of sheaves of abelian group on X . Then:

1. If $\mathcal{F}$ is flabby then

$$0 \longrightarrow \mathcal{F}(U) \xrightarrow{\varphi(U)} \mathcal{G}(U) \xrightarrow{\psi(U)} \mathcal{H}(U) \longrightarrow 0$$

is exact for any $U \subset X$ open set;

2. If $\mathcal{F}$ and $\mathcal{G}$ are flabby, then also $\mathcal{H}$ is flabby.

Definition 2.7. Given a topological space X and a sheaf $\mathcal{F}$ of abelian group on it; we say that $\mathcal{F}$ is acyclic if $H^i(X, \mathcal{F}) = 0$ for any $i \in \mathbb{Z}_{\geq 1}$

Proposition 2.8. Let X be a topological space and $\mathcal{F}$ a flabby sheaf of abelian group on X , then $\mathcal{F}$ is acyclic.

Proof. since the category of sheaves of abelian groups on a topological space X has enough injectives, we can embed $\mathcal{F}$ in an injective sheaf $\mathcal{I}$, and let be $\mathcal{G} := \mathcal{I}/\mathcal{F}$ ⁵. By the lemma above, in the exact sequence

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{I} \longrightarrow \mathcal{G} \longrightarrow 0$$

we have that, since $\mathcal{F}$ and $\mathcal{I}$ are flabby, also $\mathcal{G}$ is flabby.

We can also consider the following long exact sequence

$$0 \longrightarrow \mathcal{F}(X) \longrightarrow \mathcal{I}(X) \longrightarrow \mathcal{G}(X) \longrightarrow H^1(X, \mathcal{F}) \longrightarrow H^1(X, \mathcal{I}) \longrightarrow \dots$$

Form the first sequence and by the lemma above we have that the map $\mathcal{I}(X) \rightarrow \mathcal{G}(X)$ is surjective. Moreover, since $\mathcal{I}$ is injective, we have that $H^1(X, \mathcal{I}) = 0$; then by the exactness we have that $H^1(X, \mathcal{F}) = 0$. Now we can deal the general case by induction: let $i \geq 1$, from the above long exact sequence we have that $H^1(X, \mathcal{G}) \cong H^2(X, \mathcal{F})$. Since $\mathcal{G}$ is flabby, by the induction hypothesis, we conclude that $H^1(X, \mathcal{G}) = 0$. Then $\mathcal{F}$ is acyclic. $\square$

Lemma 2.9. Let $\mathcal{C}$ be an abelian category with enough injective, and $F : \mathcal{C} \rightarrow \mathcal{D}$ a left exact functor. Assume that A is an object of $\mathcal{C}$ for which there exists a resolution

$$A \rightarrow B^0 \rightarrow B^1 \rightarrow B^2 \rightarrow \dots$$

with $R^i F(B^j) = 0$ for all $i > 0$ and $j \geq 0$. Then $R^i F(A) \cong H^i F(B^\bullet)$.

⁵Since $\mathcal{F}$ is embedded in $\mathcal{I}$ there exists a monomorphism $f : \mathcal{F} \rightarrow \mathcal{I}$, then we have that the quotient is the cokernel of the map; i.e. $\mathcal{I}/\mathcal{F} = \text{cokernel}(f)$.

From the point of view of sheaves, $\mathcal{G}$ is the fascification of the presheaf such that for any $U \subset X$ open set maps $\mathcal{G}(U) = \mathcal{I}(U)/\mathcal{F}(U)$ where the quotient is well defined since the f is a monomorphism, hence is an injective map on the stalks and on the sections; then we can see $\mathcal{F}(U)$ as a subgroup of $\mathcal{I}(U)$

Proof. We can split the resolution of A in short exact sequences

$$0 \rightarrow A \rightarrow B^0 \rightarrow K^0 \rightarrow 0, \quad 0 \rightarrow K^0 \rightarrow B^1 \rightarrow K^1 \rightarrow 0, \dots$$

where $K^0 := \text{coker}(A \rightarrow B^0)$ and $K^j := \text{coker}(K^{j-1} \rightarrow B^j)$. From the property of the cokernel it is clear that they are all exacts ⁶. Since F is left exact, the first sequence gives an exact sequence

$$0 \rightarrow F(A) \rightarrow F(B^0) \rightarrow F(K^0) \rightarrow R^1F(A) \rightarrow R^1F(B^0) \rightarrow R^1F(K^0) \rightarrow R^2F(A) \rightarrow \dots$$

and since $R^1F(B^0) = 0$ we have that the sequence

$$0 \rightarrow F(A) \rightarrow F(B^0) \rightarrow F(K^0) \rightarrow R^1F(A) \rightarrow 0$$

is a short exact sequence. Then we have that

$$\begin{aligned} R^1F(A) &\cong \text{coker}(F(B^0) \rightarrow F(K^0)) \cong \\ &\cong \text{coker}(F(B^0) \rightarrow \ker(F(B^1) \rightarrow F(B^2))) = H^1F(B^\bullet) \end{aligned}$$

where we used that

$$\begin{aligned} F(K^0) &= F(\text{Coker}(A \rightarrow B^0)) = F(\text{Coker}(\text{Ker}(B^0 \rightarrow B^1))) \cong \\ &\cong F(\text{Ker}(\text{Coker}(B^0 \rightarrow B^1))) = F(\text{Ker}(B^1 \rightarrow B^2)) = \\ &= \text{Ker}(F(B^1) \rightarrow F(B^2)) \end{aligned}$$

⁷ Now let $j > 0$, then from the exact sequence

$$0 \rightarrow K^{j-1} \rightarrow B^j \rightarrow K^j$$

we get the exact sequences

$$0 \rightarrow F(K^{j-1}) \rightarrow F(B^j) \rightarrow F(K^j) \rightarrow R^1F(K^{j-1}) \rightarrow R^1F(B^j) \rightarrow R^1F(K^j) \rightarrow R^2F(K^{j-1}) \rightarrow \dots$$

Since $R^iF(B^j) = 0$ for any $i > 0$, we have that

- $R^iF(K^0) \cong R^{i+1}F(A)$;
- $R^iF(K^j) \cong R^{i+1}F(K^{j-1})$;

for $i > 0$ and $j \geq 1$, and also the following exact sequence

$$0 \rightarrow F(K^{i-1}) \rightarrow F(B^i) \rightarrow F(K^i) \rightarrow R^1F(K^{i-1}) \rightarrow 0$$

⁶In the category of group we have $K^0 = \frac{B^0}{\text{Im}(A \rightarrow B^0)}$ and the map $B^0 \rightarrow K^0$ is the projection (that is surjective).

⁷The first isomorphism is the *first theorem of isomorphism* for a general category. One can see the similarity with the theorem for groups (or modules). The last equality holds since F is a functor and send zero object in zero object.

Now we can compute the cohomology, indeed

$$\begin{aligned}
R^{i+1}F(A) &\cong R^iF(K^0) \cong R^{i-1}F(K^1) \cong \dots \\
&\cong R^1F(K^{i-1}) \\
&\cong \text{Coker}(F(B^i) \rightarrow F(K^i)) \\
&= \text{Coker}(F(B^i) \rightarrow \text{Ker}(F(B^{i+1}) \rightarrow F(B^{i+2}))) \\
&= H^{i+1}F(B^\bullet)
\end{aligned}$$

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□

Theorem 2.10. Let $\mathcal{F}$ be a sheaf on a topological space X . Suppose there exists a resolution

$$\mathcal{F} \rightarrow \mathcal{G}^\bullet$$

where the $\mathcal{G}^i$ are flabby sheaves (or just acyclic). Then

$$H^i(X, \mathcal{F}) \cong H^i(\Gamma(X, \mathcal{G}^\bullet))$$

Proof. We can apply the above lemma with $F = \Gamma(X, -)$ and with $B^\bullet = G^\bullet$. The condition $R^iF(B^j) = 0$ for $i > 0$ is satisfied by 2.9. □

⁸The last equalities can be proved similarly to the case $i + 1 = 1$.

3 Sheaf of stalks of non-continuous section

Definition 3.1. Let X be a topological space and $\mathcal{F}$ a sheaf of abelian group on X . The **sheaf of stalks of non-continuous section** of $\mathcal{F}$ is the sheaf $\mathcal{D}(\mathcal{F})$ defined, for any open set $U \subset X$, by

$$\mathcal{D}(\mathcal{F})(U) = \{s : U \rightarrow \bigsqcup_{x \in U} \mathcal{F}_x \mid s(x) \in \mathcal{F}_x \forall x \in U\}$$

where, if $s \in \mathcal{D}(\mathcal{F})(U)$, the restriction is

$$\rho_V^U(s) = s|_V : V \rightarrow \bigsqcup_{x \in U} \mathcal{F}_x$$

Proposition 3.2. The following statements are true:

1. exists a canonical injective morphism $\mathcal{F} \rightarrow \mathcal{D}(\mathcal{F})$;
2. $\mathcal{D}(\mathcal{F})$ is a flabby sheaf;
3. let $\mathcal{D}^{-1}(\mathcal{F}) = \mathcal{F}$, $\mathcal{D}^0(\mathcal{F}) = \mathcal{D}(\mathcal{F})$ and for any $j \geq 1$

$$\mathcal{D}^{j+1}(\mathcal{F}) = \mathcal{D}(\text{Coker}[\mathcal{D}^{j-1}(\mathcal{F}) \rightarrow \mathcal{D}^j(\mathcal{F})])^9$$

The complex $\mathcal{D}^\bullet(\mathcal{F})$ defined by

$$\mathcal{D}^\bullet(\mathcal{F}) : \mathcal{F} \rightarrow \mathcal{D}(\mathcal{F}) \rightarrow \mathcal{D}^1(\mathcal{F}) \rightarrow \dots$$

is resolution of flabby sheaves for $\mathcal{F}$.

Proof.

1. we recall that given a $\sigma \in \mathcal{F}(U)$ we can define a function $s_\sigma : U \rightarrow \bigcup_{x \in U} \mathcal{F}_x$ such that $s_\sigma(x) = \sigma_x$. Now we can define a morphism of sheaves in the following way: for any U open subset of X

$$\begin{array}{ccc} \psi : \mathcal{F}(U) & \rightarrow & \mathcal{D}(\mathcal{F})(U) \\ \sigma & \mapsto & s_\sigma \end{array}$$

since $\mathcal{F}$ is a sheaf we have that $\mathcal{F}$ is injective if and only if every stalk of $\mathcal{F}$ is injective if and only if every section $\mathcal{F}(U)$ is injective. Then we can prove the injectivity on the sections: if $s_\sigma = 0$, then we have that $0 = s_\sigma(x) = \sigma_x = [\sigma, U]$ for any $x \in U$. Since we have the equality for every stalk we have that $\sigma = 0$ ¹⁰ and that the map is injective;

⁹We can see the Coker in the categorical way (as we have done above), or in through the following way: given a morphism of sheaves $f : \mathcal{F} \rightarrow \mathcal{G}$ we can define the presheaf $\text{PCoker } f(U) := \text{Coker}(f(U))$. In general this is not a sheaf, but we can take the sheafification and define $\text{Coker } f := \text{PCoker}(f)^\dagger$.

¹⁰Let $x, y \in \mathcal{F}(U)$, since for any $p \in X$ $x_p = y_p$, then exists an open set $p \in W_p \subset U$ such that $\rho_{U_p}^U(x) = \rho_{U_p}^U(y)$; in this way, we have just built a cover for $U = \bigcup_{p \in X} U_p$. Let $x_{U_p} := \rho_{U_p}^U(x)$ for any $p \in X$, now for any pair of point $p, q \in X$ and for any $W \subset U_p \cap U_q$ open set we have

$$\rho_W^{U_p}(x_{U_p}) = \rho_W^{U_p}(\rho_{U_p}^U(x)) = \rho_W^U(x) = \rho_W^{U_q}(\rho_{U_q}^U(x)) = \rho_W^{U_q}(x_{U_q})$$

since $\mathcal{F}$ is a sheaf, there exists a unique $z \in \mathcal{F}(U)$ such that $\rho_{U_p}^U(z) = x_{U_p}$. Since also x is such that $\rho_{U_p}^U(x) = x_{U_p}$ we have $x = z$. We can do all the steps above with $y = x$ and get that $z = y$. Then we have $x = y$.

2. let $s \in \mathcal{D}(\mathcal{F})(U)$, then we have that

$$s : U \rightarrow \bigsqcup_{x \in U} \mathcal{F}_x$$

we can extend s to a function $\bar{s} : X \rightarrow \bigsqcup_{x \in U} \mathcal{F}_x$ such that

$$p \mapsto \begin{cases} 0 & \text{if } p \notin U \\ s(p) & \text{if } p \in U \end{cases}$$

where the function is well defined since $\mathcal{F}_x$ is a group, then there exists 0 in it. Then we have that $\rho_U^X(\bar{s}) = \bar{s}|_U = s$ and this implies that $\mathcal{D}(\mathcal{F})$ is flabby;

3. The complex is

$$0 \rightarrow \mathcal{F} \xrightarrow{i^1} \mathcal{D}(\mathcal{F}) \xrightarrow{i^2} \mathcal{D}(\text{Coker}(i^1)) \rightarrow \dots$$

where the map i^2 is the composition of the projection $\mathcal{D}(\mathcal{F}) \rightarrow \text{Coker}(i^1)$ and the injective map $\text{Coker}(i^1) \rightarrow \mathcal{D}(\text{Coker}(i^1))$ given by point 1 of this proposition; one can show that is a complex and that is exact.

□

Now we have just cooked a resolution of flabby sheaves for a sheaf $\mathcal{F}$, then we can compute the cohomology of $\mathcal{F}$ using the last theorem of the previous chapter:

$$H^i(X, \mathcal{F}) \cong H^i(\Gamma(X, \mathcal{D}(\mathcal{F})))$$